

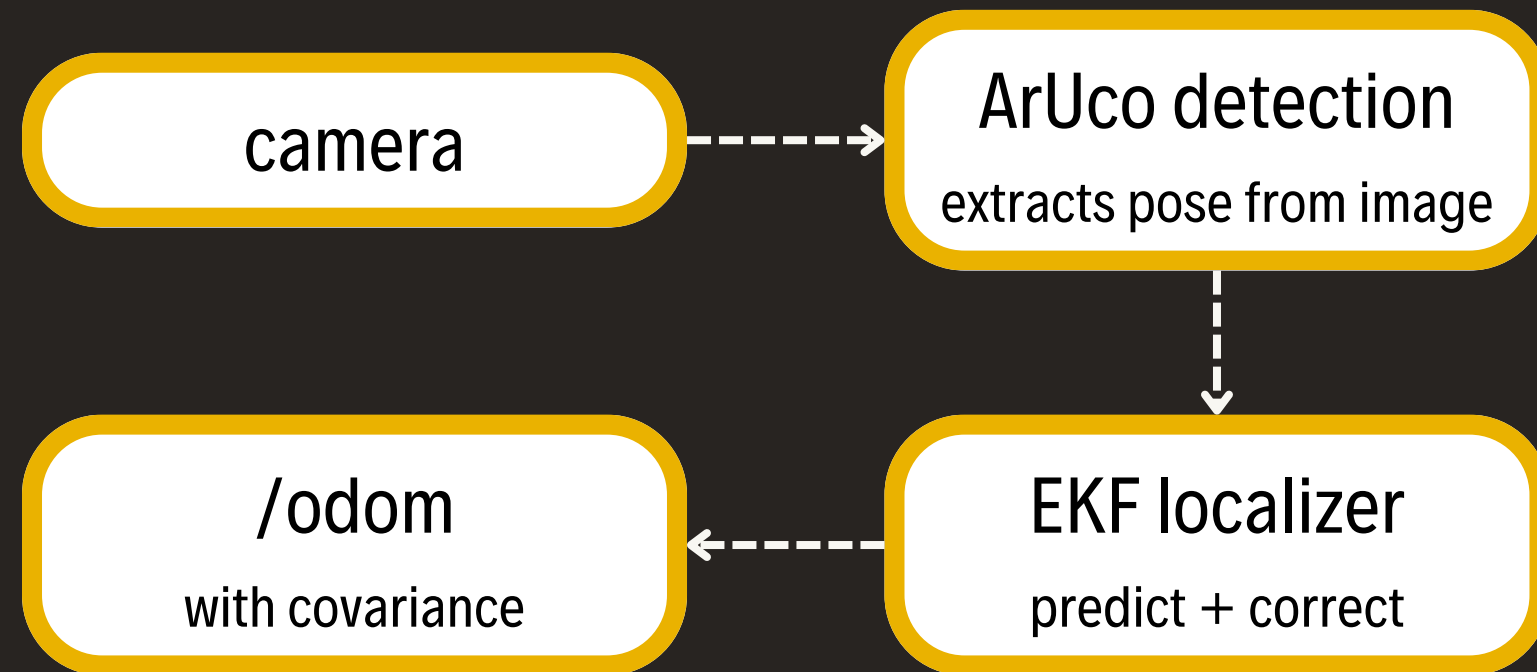
MCR2 Final Challenge: **Autonomous exploration**

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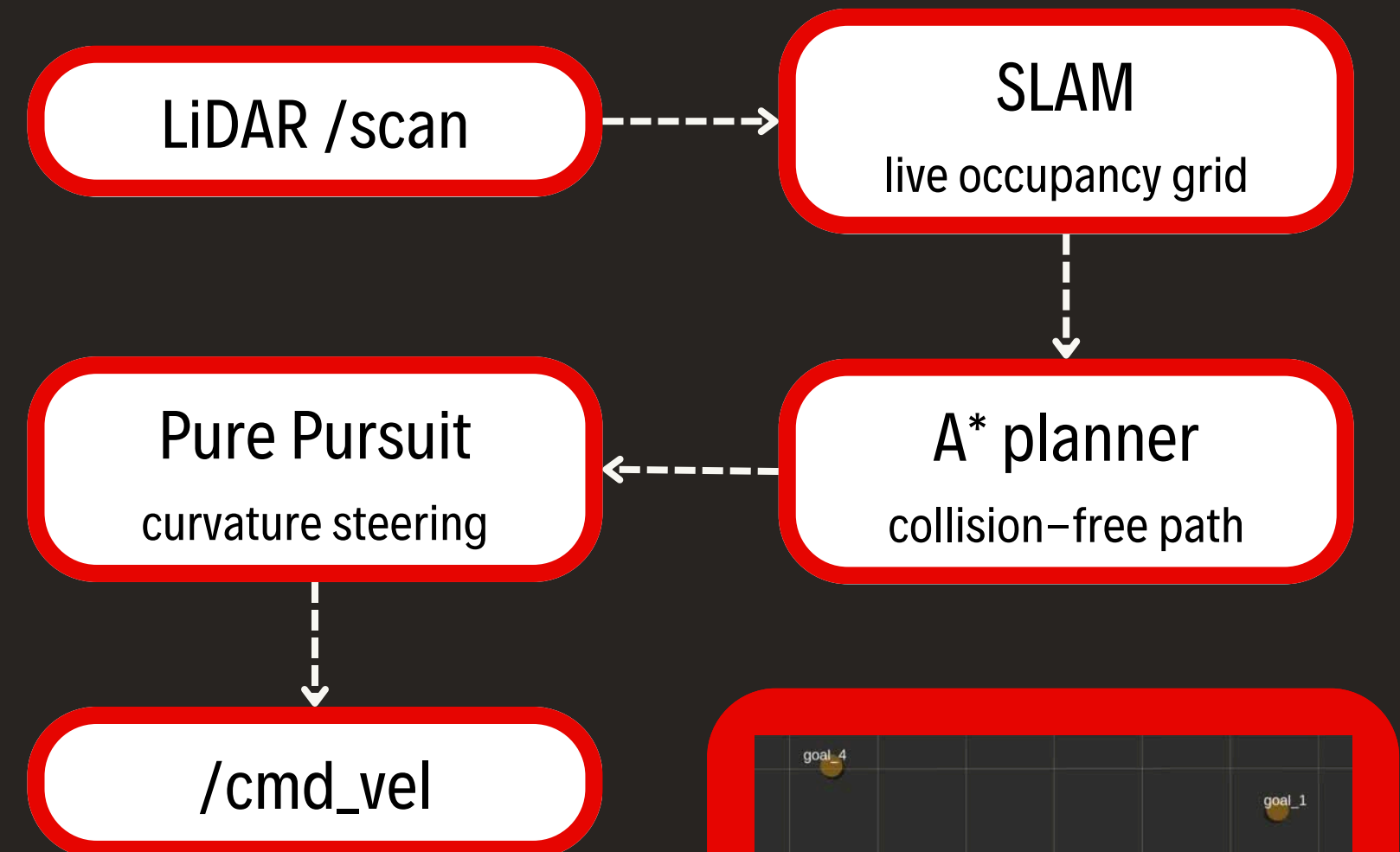
Integration of Robotics and Intelligent Systems



Part 1: Localisation

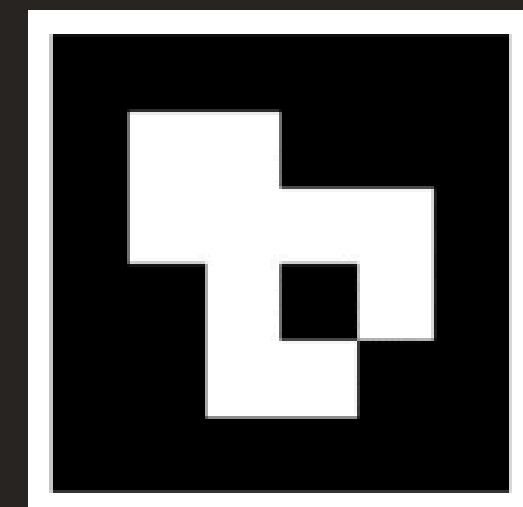
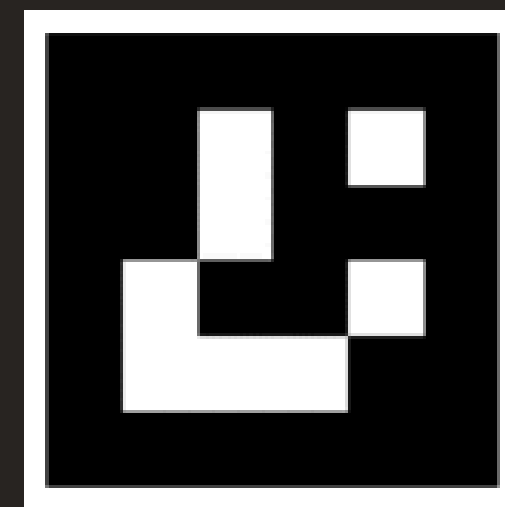
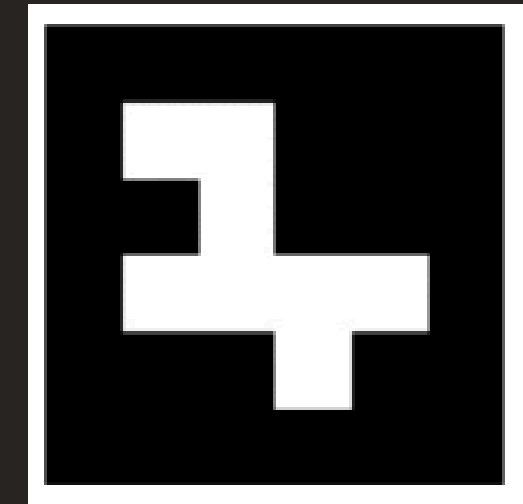
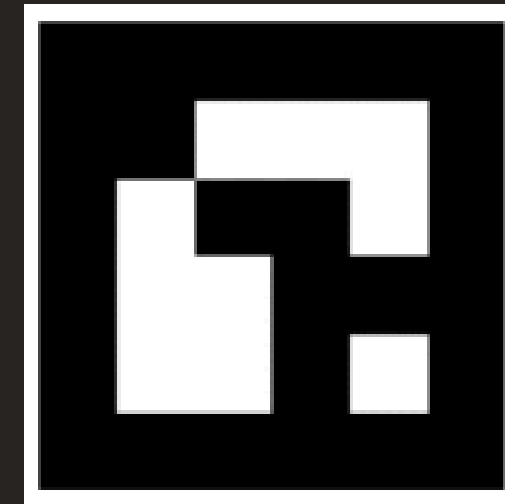
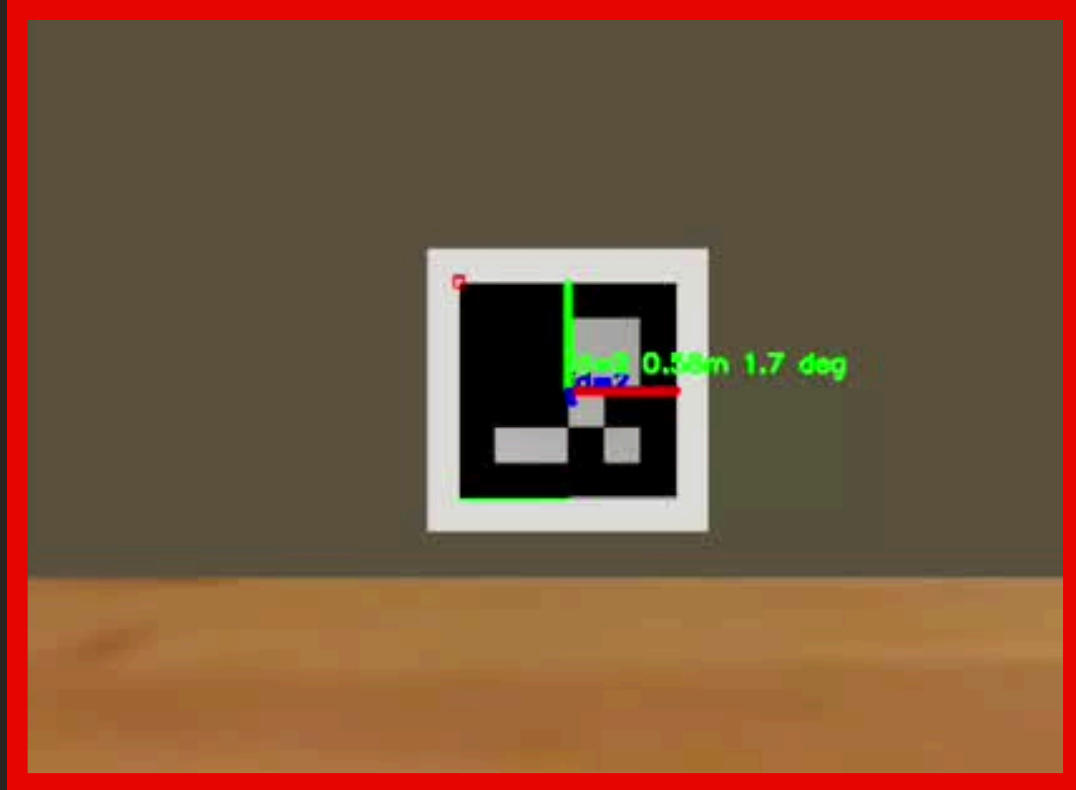


Part 2: Unknown exploration



ArUco markers

- Square visual markers with unique IDs.
- Detected automatically by a camera.
- Used to estimate position and orientation.



Camera Node

Purpose

- Detects ArUco markers from the robot camera stream.
- Estimates each marker's 3D position and orientation.
- Publishes marker poses and TF transformations.

Publishes

- Marker poses (/aruco/detections)
- TF frames (camera → marker)
- Debug image with visual annotations.

Receives camera images and intrinsic parameters.

Converts images to grayscale.

Detects ArUco markers using OpenCV.

Estimates marker pose with the PnP algorithm.

Computes position and quaternion orientation.

Kalman Filter

Prediction.

wheel odometry

$$\mathbf{s}_k = \mathbf{h}(\mathbf{s}_{k-1}, \mathbf{u}_k) + \mathbf{q}_k$$

- Always available
- Error accumulates over time
- Uncertainty grows with motion

Measurement.

camera / ArUco

$$\mathbf{z}_{i,k} = \mathbf{g}(\mathbf{m}_i, \mathbf{s}_k) + \mathbf{r}_{i,k}$$

- Only when marker visible
- More precise, no drift
- Uncertainty fixed by calibration

Covariance formula

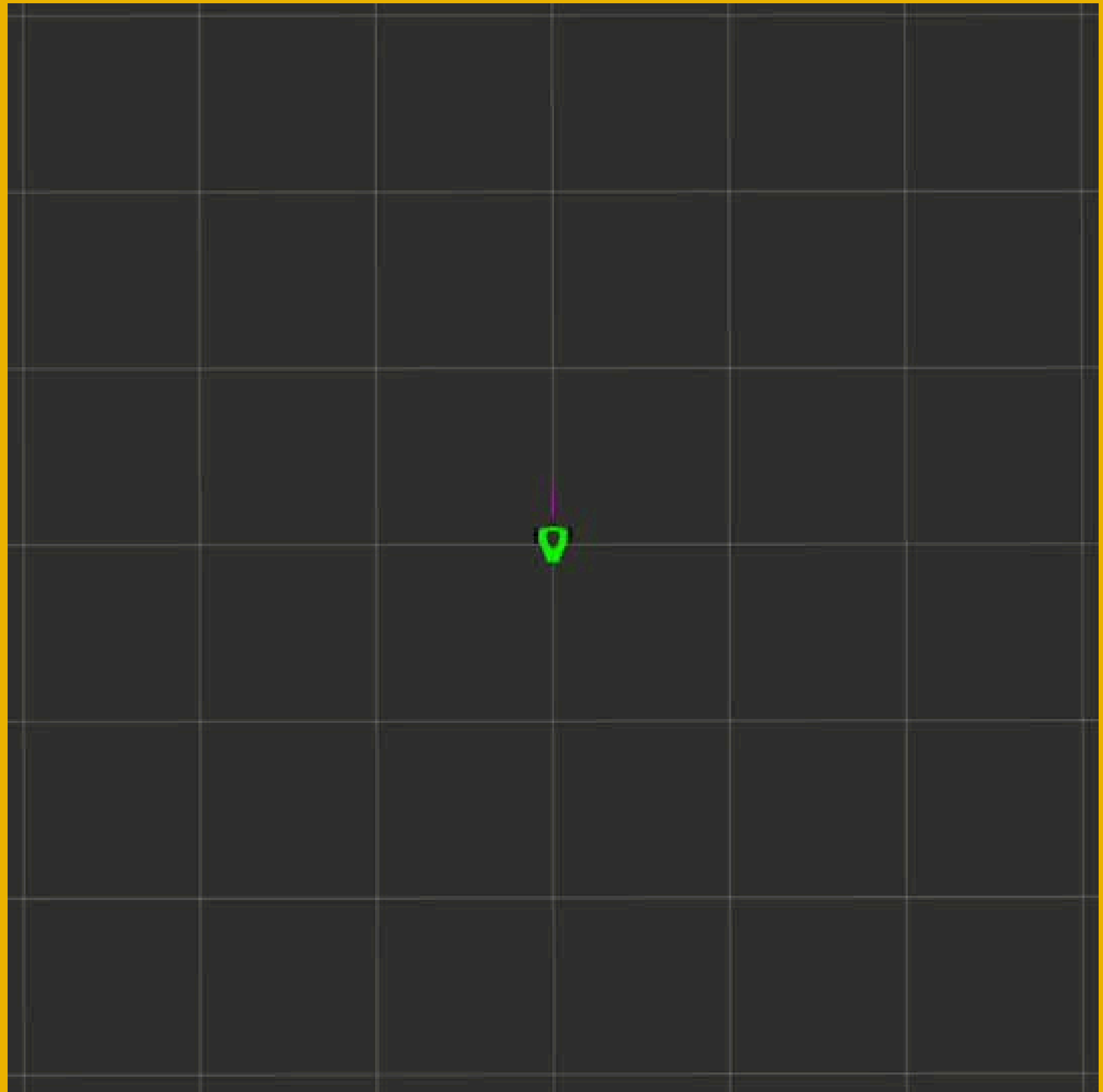
$$\Sigma_k = \mathbf{H}_k \Sigma_{k-1} \mathbf{H}_k^T + \mathbf{Q}_k$$

3x3 covariance matrix

$$\Sigma_k = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{x\theta} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{y\theta} \\ \sigma_{\theta x} & \sigma_{\theta y} & \sigma_{\theta\theta} \end{bmatrix}$$

3x3 Linear model Jacobian of the robot

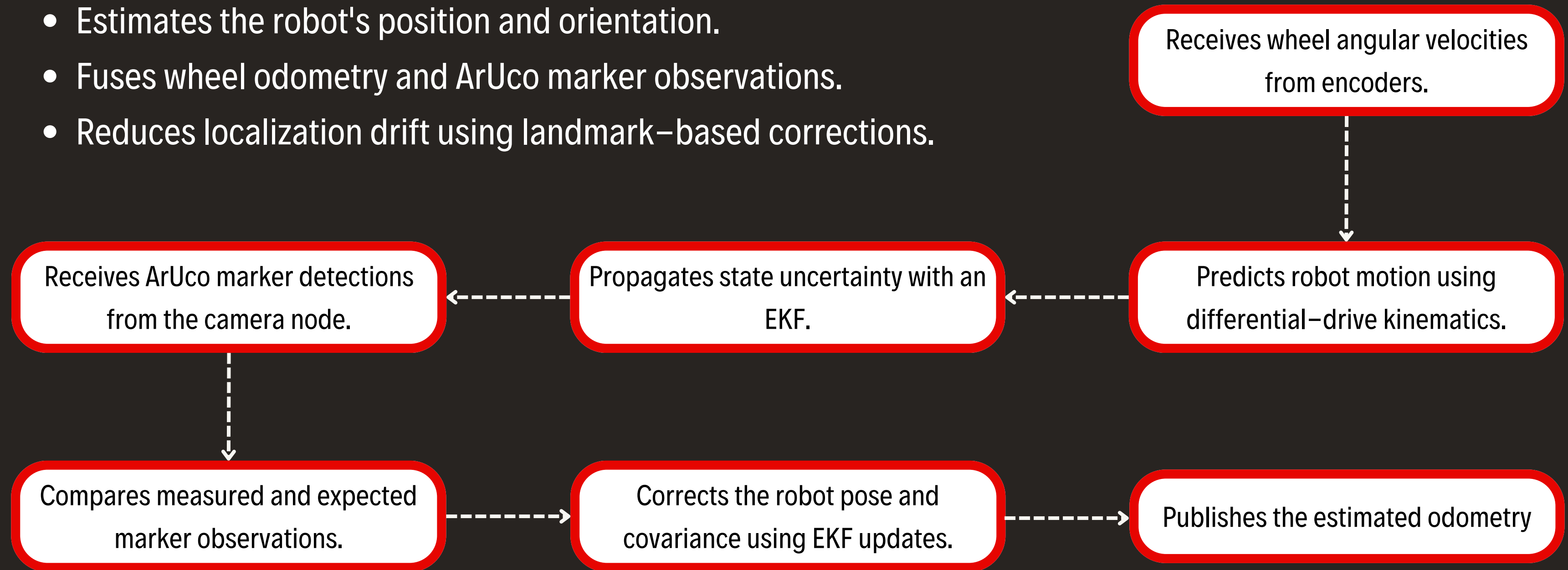
$$\mathbf{H}_k = \begin{bmatrix} 1 & 0 & -\Delta t \cdot v_k \cdot \sin(\mu_{\theta,k-1}) \\ 0 & 1 & \Delta t \cdot v_k \cdot \cos(\mu_{\theta,k-1}) \\ 0 & 0 & 1 \end{bmatrix}$$



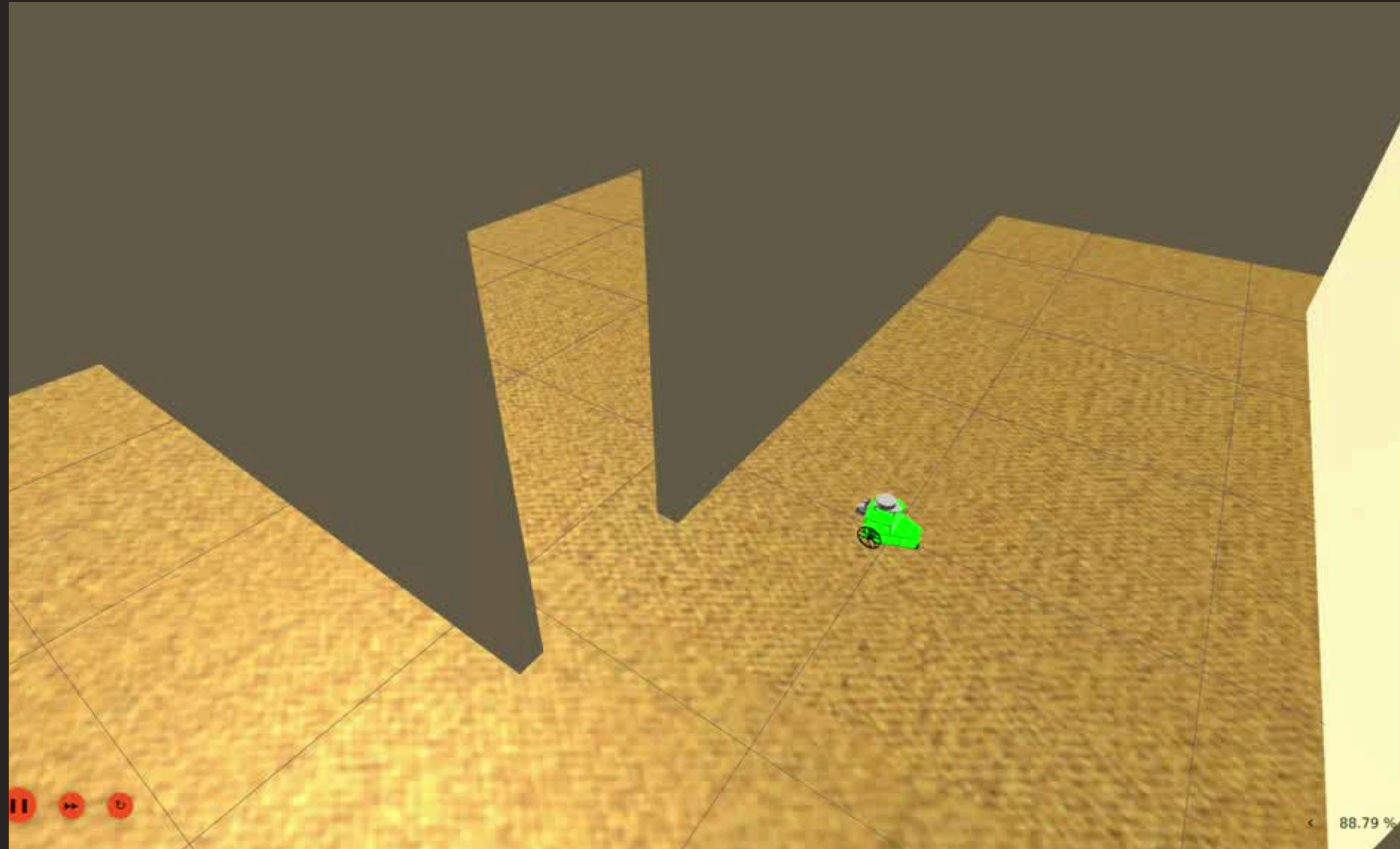
Localisation Node

Purpose

- Estimates the robot's position and orientation.
- Fuses wheel odometry and ArUco marker observations.
- Reduces localization drift using landmark-based corrections.



Mission Node



Loads waypoints from a configuration file.



Waits for the mission start delay.



Publishes the active waypoint as /goal_pose.



Monitors the /goal_reached signal.



Advances to the next waypoint when the current one is reached.

Planner Node

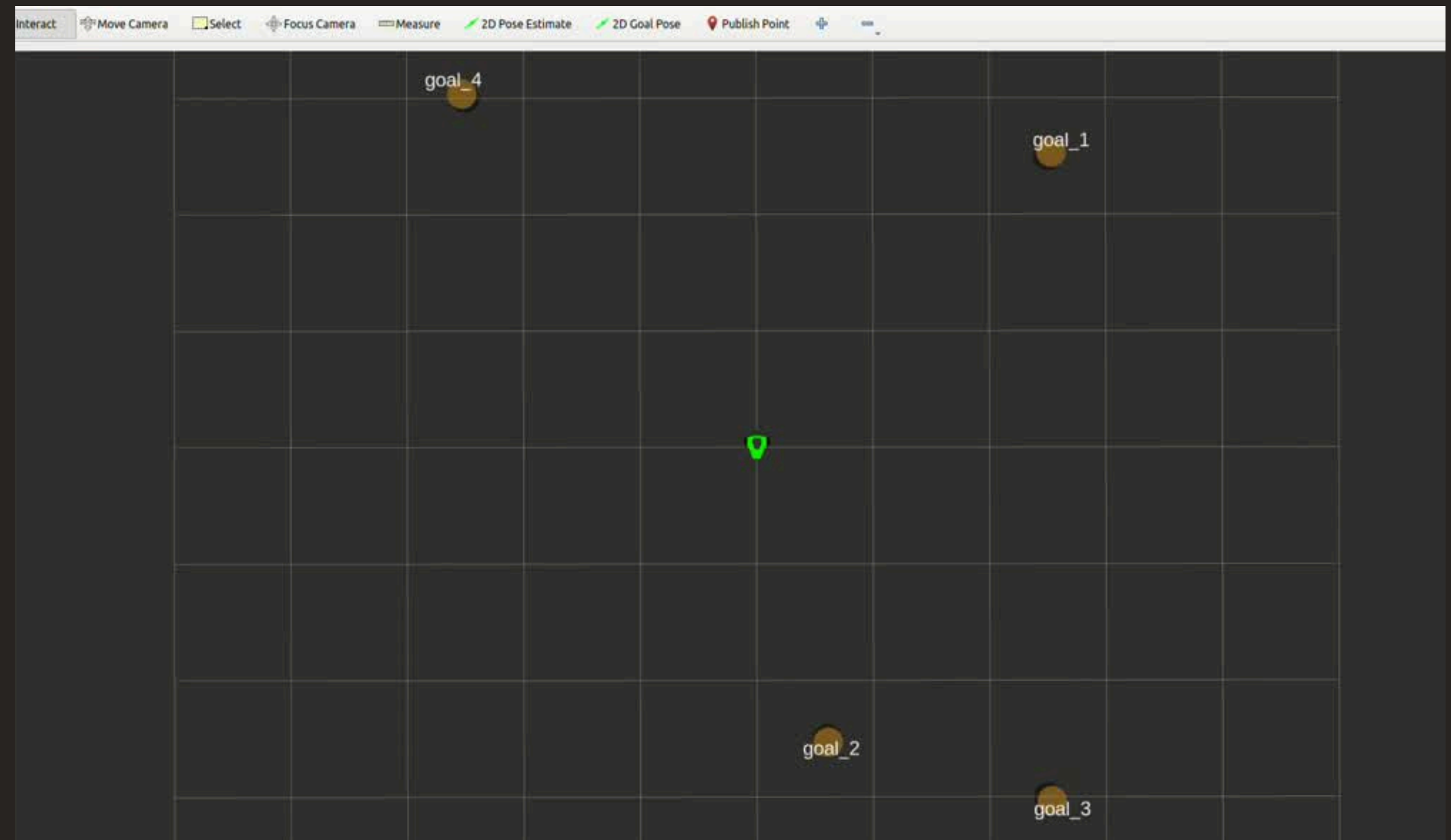
Receives the SLAM occupancy grid
/map.

Receives the current goal /goal_pose.

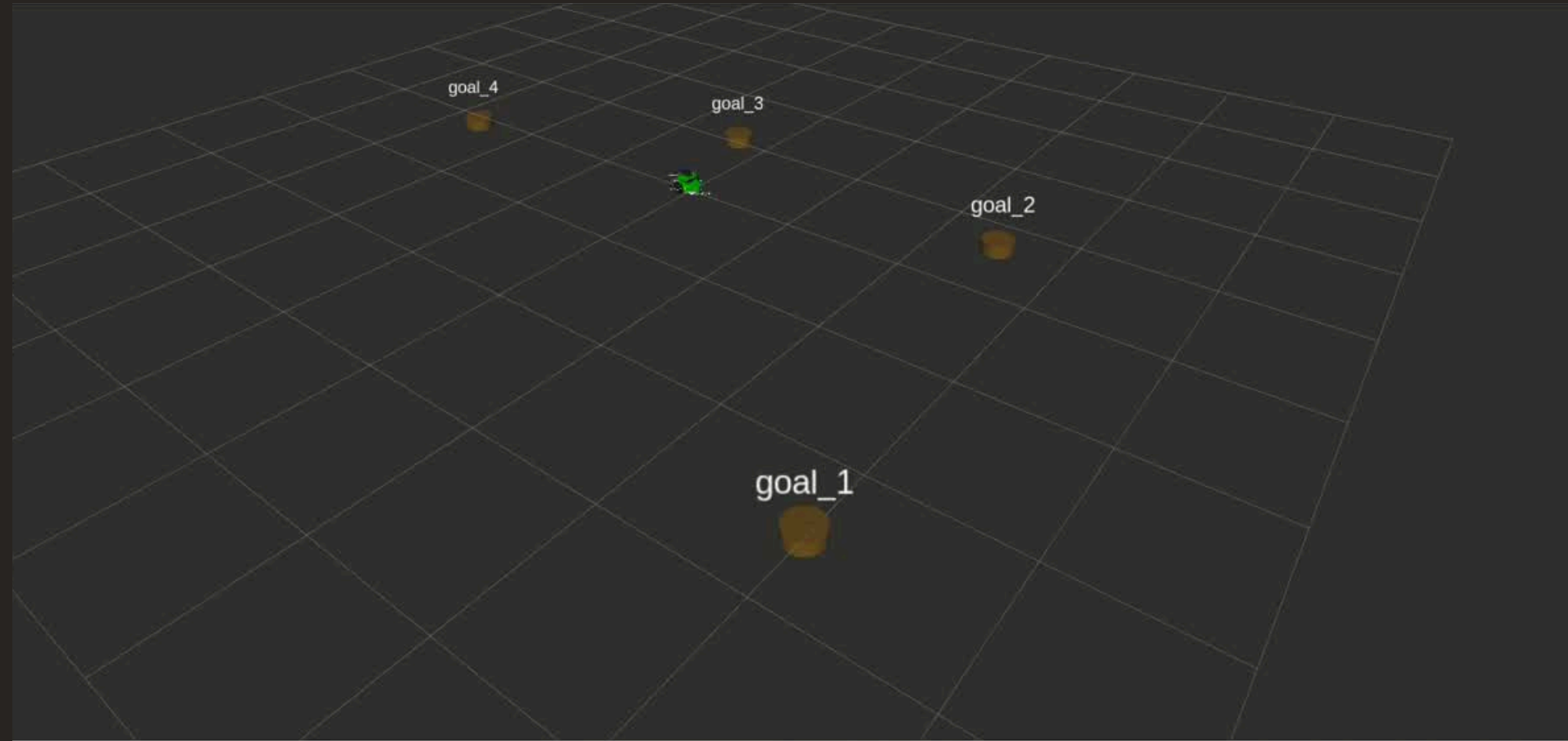
Inflates obstacles according to the
robot size.

Updates free space using live LiDAR
data.

Runs the A* search algorithm.



Controller Node



Detects when the goal is reached and publishes a completion signal.

Generates linear and angular velocity commands /cmd_vel.

Applies the Pure Pursuit algorithm to calculate steering curvature.

Receives the planned path /plan.

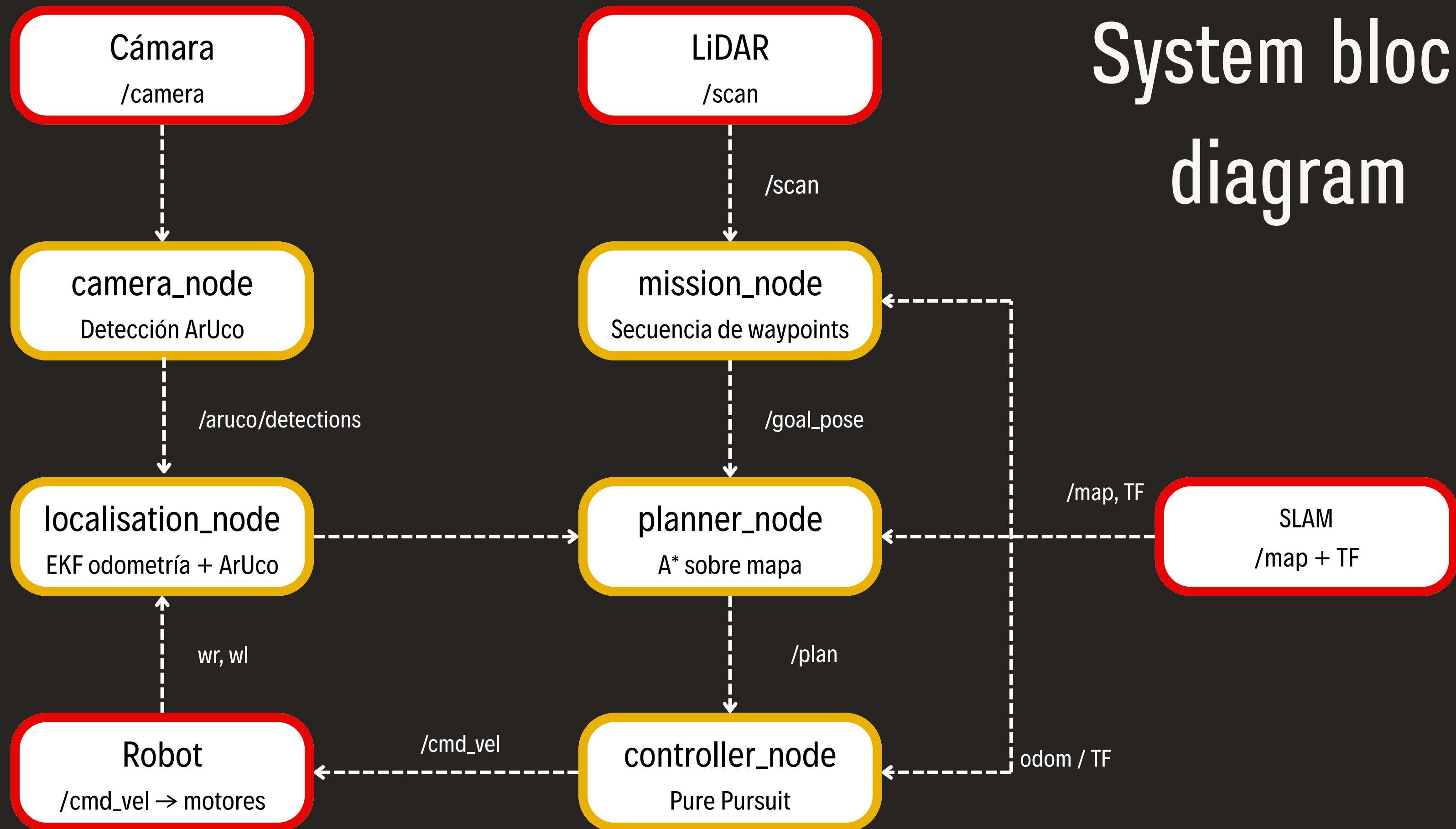
Obtains the robot pose from TF.

Selects a lookahead point along the path.

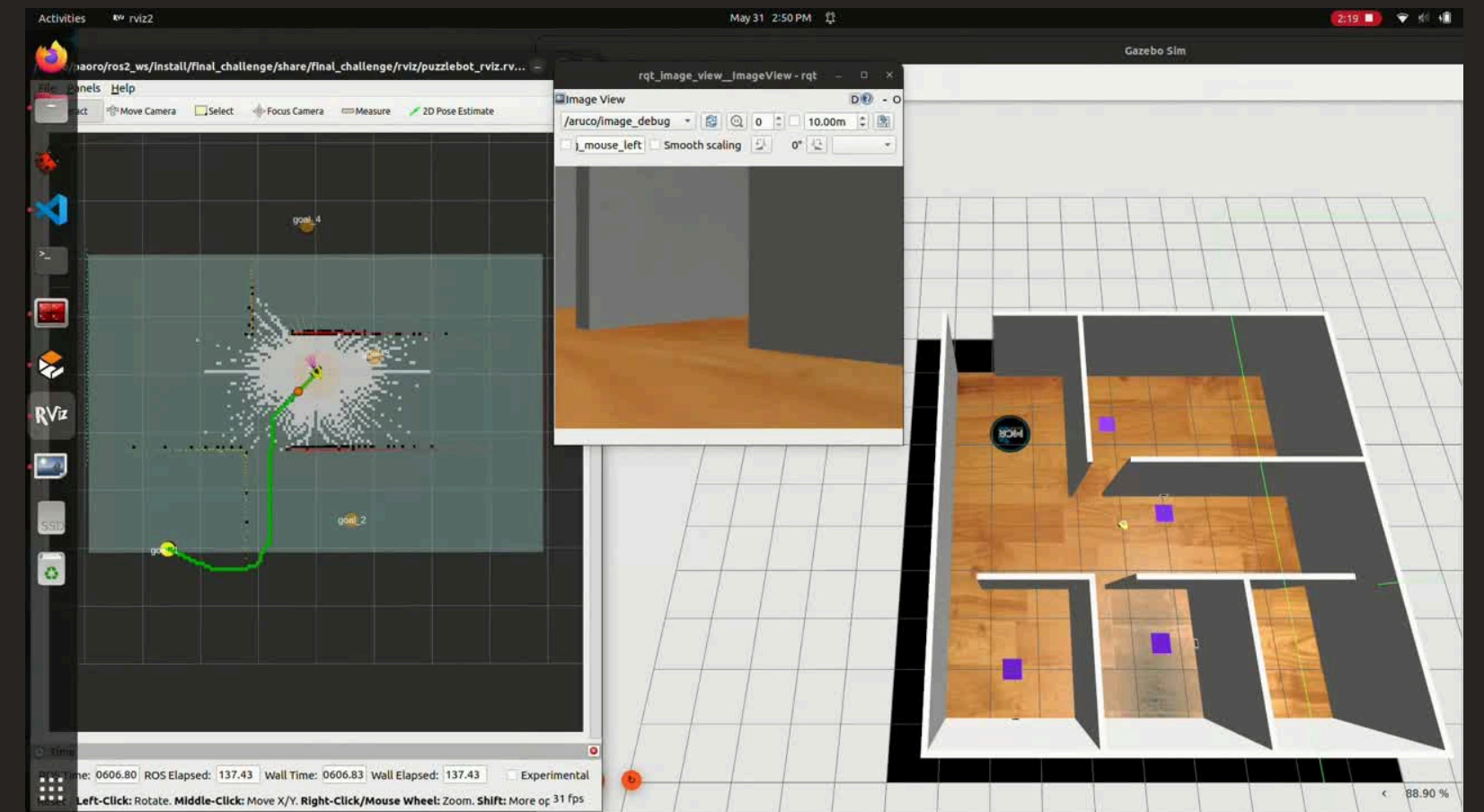
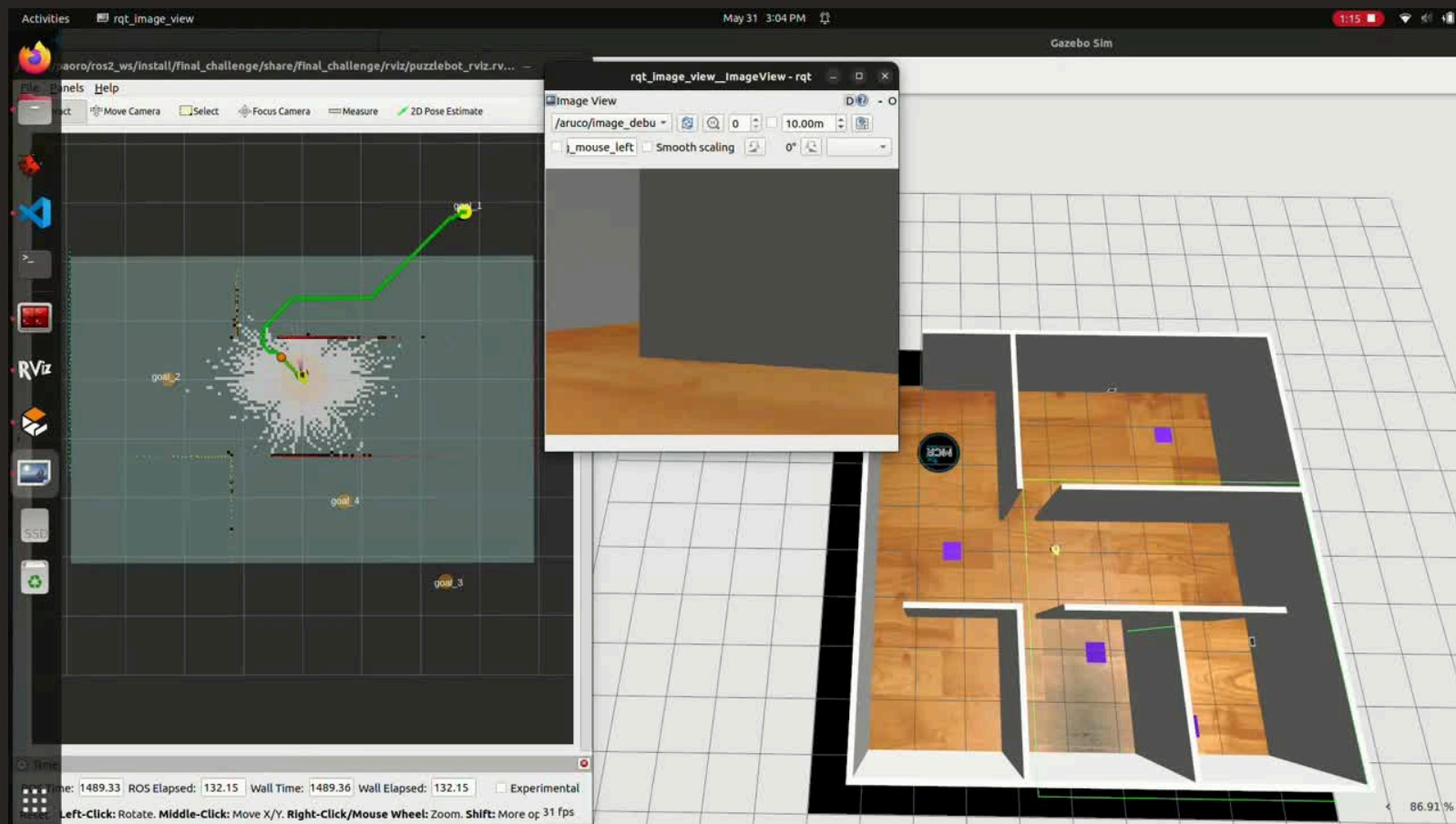
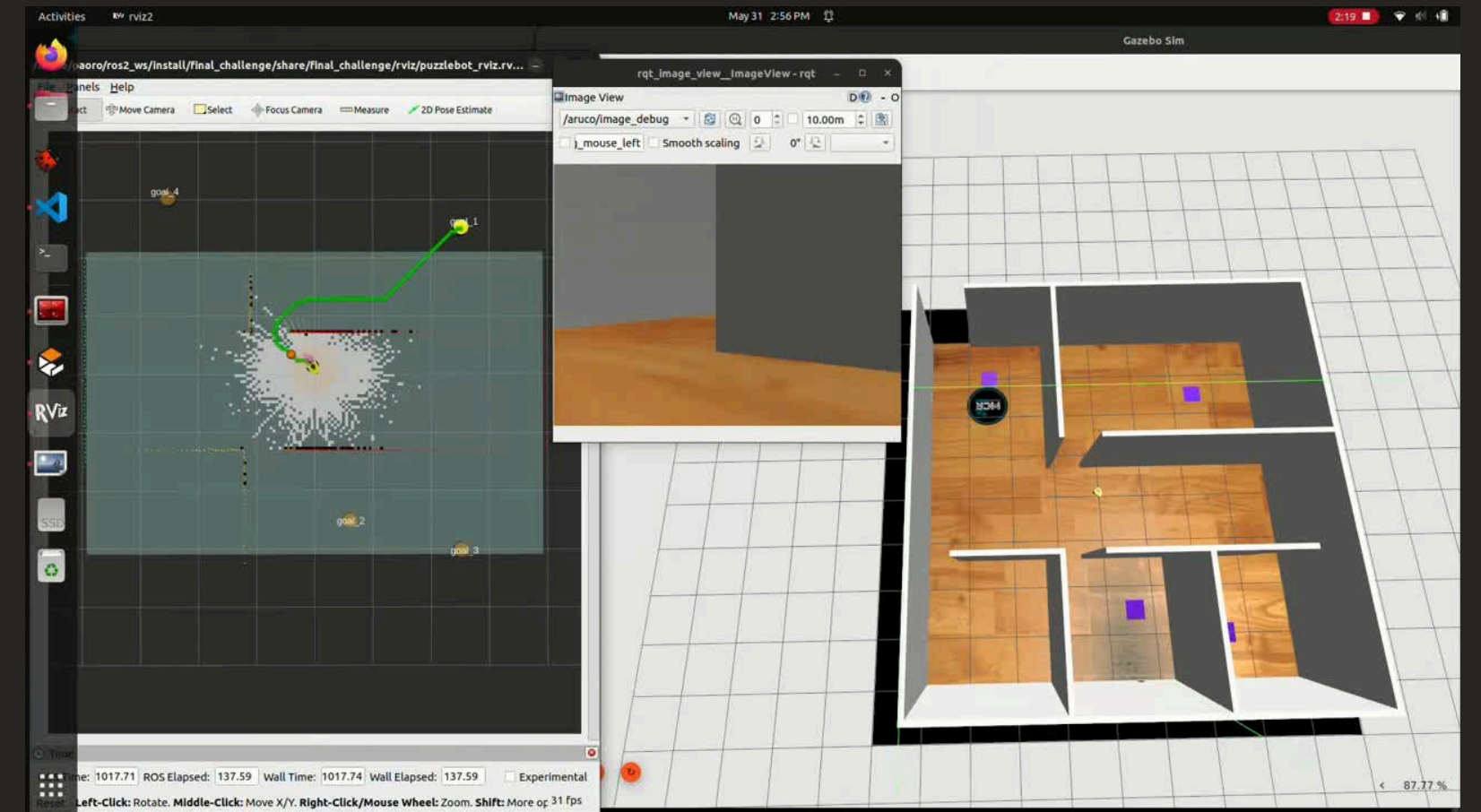
Computes the heading error to the target point.



System block diagram



Final Results



Thank you.

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github.com/PaolaRojas24/guns-n-ROSeS/tree/main/Week7